SOCIOL 723: Problem Set 1

***Assigned*: September 1 • *Due*: Monday, September 14, 11:59PM**

Your name:

Instructions. Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. Show your work: for the short conceptual questions, a few clear sentences; for computation, the code and its output. You may use AI assistance, but you are responsible for understanding and being able to explain every line you submit. No question requires a derivation: everything here is a computation, an arithmetic plug-in, or a few clear sentences.

Use `set.seed(723)` at the top of each simulation so your results are reproducible. The data file is `Data/gss_earnings.rds`, the same extract used in lab.

OLS in Your Hands (25 points)

Simulate one sample of size $n = 200$ from

$$Y_i = 1 + 0.5\, X_i + \epsilon_i, \qquad X_i \sim N(0,1), \quad \epsilon_i \sim N(0,1).$$

1. Compute the slope with sums only, no `lm()`:

$$\hat\beta_1 = \frac{\sum_i (X_i - \bar X)(Y_i - \bar Y)}{\sum_i (X_i - \bar X)^2}, \qquad \hat\beta_0 = \bar Y - \hat\beta_1 \bar X.$$

   Then confirm both against `lm(y ~ x)`. Why is $\hat\beta_1$ not exactly 0.5 even though the code contains the true model? One sentence.

2. Compute the residuals $\hat e_i = Y_i - \hat\beta_0 - \hat\beta_1 X_i$ and report `sum(e)` and `sum(e * x)`. Both should be zero to machine precision ($\approx 10^{-12}$). In one or two sentences: are these zeroes evidence that the model is right? What are they instead?

3. Rerun the regression with the *centered* regressor `x - mean(x)`. Make a small table comparing $\hat\beta_0$, $\hat\beta_1$, R^2, and the first three residuals across the two runs. One sentence: what does centering change, and what does it leave alone?

4. **GSS.** Regress `lnearn` on `educ`. Report the slope, interpret it in one sentence (remember the outcome is in logs), and produce a scatterplot with the fitted line drawn through it.

The Sampling Distribution You Never See (25 points)

In real life you get one sample. In a simulation you can run the thought experiment from lecture for real: draw sample after sample from the same population and watch what $\hat{\beta}_1$ does.

1. Using the same population as Question 1 with $n = 50$: draw 5,000 samples, save the 5,000 slopes, and plot their histogram with a normal density overlaid. Report the mean and standard deviation of the 5,000 slopes. Where is the histogram centered, and what course concept is the standard deviation you just computed?

2. For each of the 5,000 samples, also save the standard error that `summary(lm(...))` reports for the slope. Compare the *average reported standard error* to the *standard deviation of the slopes* from part 1. One or two sentences: what is each of these two numbers estimating, and why should they be close?

3. Replace the normal errors with something aggressively non-normal, for example `rexp(n) - 1` (skewed, mean zero). At $n = 50$, is the histogram of the 5,000 slopes still bell-shaped? Which theorem is doing the work, and what assumption about the *errors' distribution* did it need?

Partialling Out, Seen Once (25 points)

In lecture, the Frisch–Waugh–Lovell theorem: the coefficient on D_i in a multiple regression equals the slope from the bivariate regression of the residualized outcome on the residualized treatment,

$$\beta_1 = \frac{\mathrm{Cov}(\tilde{Y}_i, \tilde{D}_i)}{\mathrm{Var}(\tilde{D}_i)},$$

where $\tilde{D}_i$ and $\tilde{Y}_i$ are residuals from regressing D_i and Y_i on the controls.

1. **GSS.** For the model

   ```r
   lm(lnearn ~ educ + exper + I(exper^2) + female + black + wordsum + pareduc,
      data = gss)
   ```

   residualize `educ` and `lnearn` on all the other regressors, run the bivariate regression of the residualized outcome on the residualized treatment, and confirm that the slope matches the multiple-regression coefficient on `educ` to at least six decimal places.

2. Produce the added-variable plot (residualized `lnearn` against residualized `educ`, with the fitted line).

3. A colleague adds a control that is very highly correlated with `educ`, watches the standard error on `educ` explode, and concludes something has gone wrong. In two or three sentences, using the idea that the standard error runs on the *residualized* variation in the treatment, explain why nothing is wrong, and what tradeoff the colleague is actually facing.

Standard Errors, Various (25 points)

Simulate one sample of size $n = 200$ from the heteroskedastic population

$$Y_i = 1 + 0.5\,X_i + \epsilon_i, \qquad X_i \sim N(0,1), \quad \epsilon_i \sim N\big(0,\ (0.5\,e^{0.8X_i})^2\big).$$

1. Plot the residuals against X_i and describe the picture in one sentence. Which assumption is violated, and what does it break: $\hat{\beta}_1$, or the classical standard error?
2. Compute the robust standard error *by hand*, directly from the scalar sandwich with $d_i = X_i - \bar{X}$:

   $$\widehat{SE}(\hat{\beta}_1) = \sqrt{\frac{\sum_i d_i^2\,\hat{e}_i^2}{\big(\sum_i d_i^2\big)^2}},$$

   and confirm it against `sandwich::vcovHC(fit, type = "HC0")`; report it beside the classical standard error.
3. **Coverage.** Over 2,000 replications of this population, build the nominal 95% confidence interval for β_1 three ways (classical, HC0, and HC3) and record how often each interval contains the truth 0.5. Report the three coverage rates and comment in two sentences.
4. **Clustered, by hand.** For the GSS bivariate model `lnearn ~ educ`, with d_i the deviation of `educ` from its mean: form each sampling cluster's total $S_g = \sum_{i\in g} d_i\hat{e}_i$, one per `psu`, and square after summing:

   $$\widehat{SE}_{\mathrm{CR}}(\hat{\beta}_1) = \sqrt{\frac{\sum_g S_g^2}{\big(\sum_i d_i^2\big)^2}},$$

   and confirm it against `sandwich::vcovCL(fit, cluster = ~ psu, type = "HC0", cadjust = FALSE)`.
5. A referee writes: "You should use robust standard errors, since the key regressor is likely endogenous (i.e., you have omitted variables that will bias your main estimate)." Respond in two sentences.